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Multi-aspect data analysis for investigating human computation mechanism

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Abstract

In the paper, we present a multi-aspect data analysis approach for investigating human computation mechanism. Multi-aspect analysis in multiple human brain data sources is an important methodology in Brain Informatics, which emphasizes on a *systematic* way for investigating human information processing mechanisms, including measuring, collecting, modeling, transforming, managing, and mining multiple human brain data obtained from various cognitive experiments by using powerful equipments, such as fMRI and EEG. After giving an outline of Brain Informatics methodology, we describe how to design cognitive experiments of mental arithmetic task with multiple difficulty levels for obtaining multiple EEG and fMRI data sources, and how to analyze such data for investigating the spatiotemporal characteristics and flow of human computation processing. Such an investigation can be regarded as a case study using Brain Informatics methodology. Experimental results show the usefulness of our approach.
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Keywords: Multi-aspect data analysis; Brain informatics methodology; Human computation mechanism; EEG and fMRI

1. Introduction

Multi-aspect analysis in multiple human brain data sources is an important methodology in Brain Informatics. Brain Informatics (BI) attempts to understand human intelligence in depth, towards a long-term, holistic view to uncover the principles and mechanisms underlying human information processing system (HIPS). BI emphasizes on a *systematic* approach for investigating human information processing mechanisms, including measuring, collecting, modeling, transforming, managing, and mining multiple human brain data obtained from various cognitive experiments. The future of BI will be shaped by our ability to do large-scale analysis and simulation of human brain data. The key issues are how to design the psychological and physiological experiments for systematically obtaining various data from HIPS, as well as how to manage and analyze such data from multiple aspects for discovering new models of HIPS. Although several human-expert centric tools such as SPM (MEDx) have been developed for cleaning, normalizing and visualizing the fMRI images, researchers have also been studying how the fMRI images can be automatically analyzed and understood by using data mining and statistical learning techniques (Megalooikonomou & Herskovits, 2001; Mitchell et al., 2004; Sommer & Wichert, 2003; Zhong, Wu, Nakamaru, Ohshima, & Mizuhara, 2004). We are concerned with how to extract significant features from multiple brain data measured by using fMRI and EEG in preparation for multi-aspect analysis and simulation.

In the paper, we present a multi-aspect data analysis approach for investigating human computation mechanism with multiple difficulty levels. Such an investigation can be regarded as a case study using Brain Informatics

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methodology. We show that discovering new knowledge and models of human computation mechanism requires not only individual data obtained from a single measuring method, but also methods for multiple data sources measuring and multi-aspect data analysis. In this work, we focus on investigating the spatiotemporal characteristics and flow of human computation processing. More specifically, at the current stage, we want to understand: how a specific part of the brain operates in a specific time, how the operations change over time, and how the activated areas work cooperatively to implement a whole human computation mechanism.

The rest of the paper is organized as follows. Section 2 provides an outline of the BI Methodology. Section 3 explains how to design the experiments of ERP/fMRI mental arithmetic tasks with multiple difficulty levels standing on BI Methodology. Section 4 describes multi-aspect data analysis in ERP and fMRI based multiple data sources for investigating human computation mechanism. Section 5 discusses how to combine the results obtained by each analysis method separately, stated in Section 4, towards a more whole explanation and understanding. Finally, Section 6 gives concluding remarks.

2. Brain informatics methodology

As an interdisciplinary study, Brain Informatics (BI) combines results from experimental, computational, cognitive neuroscience and advanced Web intelligence (WI) centric information technologies. BI can be regarded as brain sciences in the WI centric IT age (Zhong, 2005, 2007; Zhong et al., 2007), which is proposing to study human brain from the viewpoint of informatics (i.e., human brain is an information processing system) and use informatics (i.e., WI centric information technologies) to support brain science studies. BI pursues a holistic understanding of human intelligence through a systematic approach to brain research. BI regards the human brain as an information processing system and emphasizes cognitive experiments to understand its mechanisms for managing and analyzing data. Such systematic study includes the following four main research issues:

- Systematic investigation of human thinking centric mechanisms;
- systematic design of cognitive experiments;
- systematic human brain data management; and,
- systematic human brain data analysis.

The capabilities of human intelligence can be broadly divided into two main aspects: perception and thinking. In cognitive neuroscience, although many advanced results with respect to "perception oriented" study have been obtained, only a few of preliminary, separated studies with respect to "thinking oriented" and/or a more whole information process have been reported (Gazzaniga, 2004). Systematic investigation of human thinking centric mechanisms is based on the observation for Web intelligence research needs and the state-of-the-art cognitive neuroscience (Zhong, 2006; Zhong et al., 2007).

Systematic design of cognitive experiments is with respect to how to design the psychological and physiological experiments for obtaining various data from HIPS, in a systematic way. In order to discover new knowledge and models of human information processing activities, not only single data source obtained from a single measuring method, but also multiple data sources from various practical measuring methods, such as fMRI and EEG, are required. Furthermore, by systematical design of cognitive experiments in BI methodology, the data obtained from a cognitive experiment and/or a set of cognitive experiments expect to be used for multi-purpose.

Systematic human brain data management is based on a conceptual brain data model, which represents functional relationships among multiple human brain data sources, with respect to all major aspects and capabilities of HIPS for systematic investigation and understanding of human intelligence. This kind of data representation is with multi-level by modeling, abstracting and transforming for multi-aspect analysis and simulation.

Systematic human brain data analysis is concerned with how to extract significant patterns/features from multiple brain data sources measured by using fMRI and EEG in preparation for multi-aspect data analysis by combining various data mining methods with reasoning (Megalooikonomou & Herskovits, 2001; Mitchell et al., 2004; Sommer & Wichert, 2003; Sternberg, Lautrey, & Lubart, 2003; Zhong, 2007). Multi-aspect data analysis can mine several kinds of hypotheses from different data sources. Furthermore, ACT-R, the cognitive architecture, is used to simulate/predict human cognitive behavior and the structure of brain to achieve the function of adaptive cognition in a system level (Anderson et al., 2004, Anderson, Qin, Yung, & Carter, 2007). Thus, the framework provides a combination of analysis and simulation to understand human intelligence in depth.

Fig. 1 shows an investigation flow based on BI methodology, in which various tools can be cooperatively used in the multi-step process for experimental design, pre-processing (data extraction, modeling and transformation), multi-aspect data mining and post-processing. In the experimental design part, various tasks of experiments with multiple difficulty levels are designed by a systematic way for multi-purpose investigation of human thinking and/or perception centric cognitive functions. In the data mining part, various methods for data processing, mining, reasoning, and simulation are deployed on a multi-phase process for performing multi-aspect analysis as well as multi-level conceptual abstraction and learning, which aims at discovering useful knowledge to understand human intelligence in depth.

3. The experiment of mental arithmetic task with multiple difficulty levels

Fig. 2 gives a computation processing from the macro viewpoint, which consists of several component functions

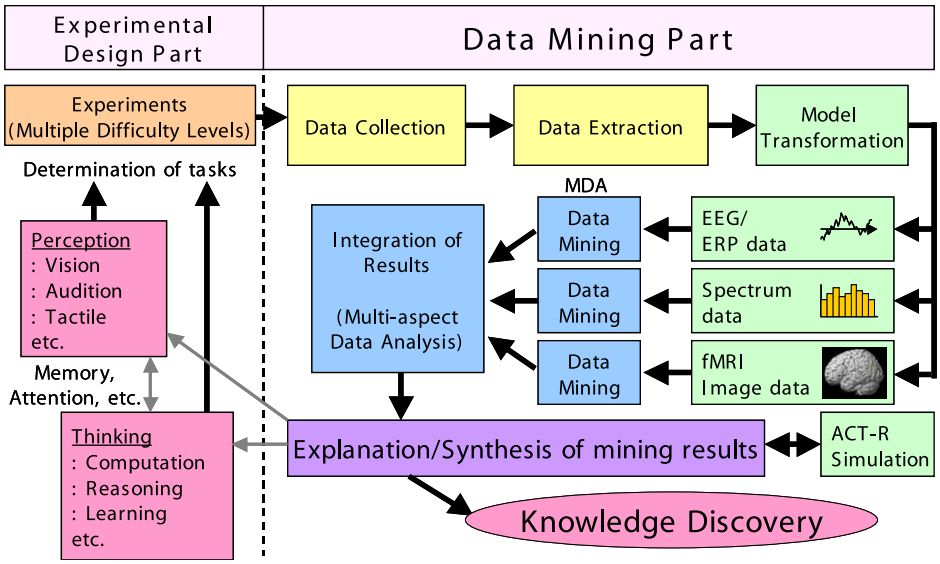


Fig. 1. An investigation flow based on BI methodology.

of human computation mechanism, such as attention, interpretation, short-term memory, understanding of work, computation, and checking. As mentioned above, based on the BI methodology, the data obtained from a cognitive experiment and/or a set of cognitive experiments expect to be used for multi-purpose. For example, it is possible that our experiments meet the following requirements: investigating the mechanisms of human visual and auditory systems, computation, problem-solving (i.e. the computation processing is regarded as an example of problem-solving processes), and spatiotemporal characteristics and flow of HIPS in general. Furthermore, we set multiple difficulty levels with respect to a series of computation tasks in a set of experiments.

3.1. Outline of experiments

The task of our cognitive experiments is to show a numerical calculation problem to a human subject, and to ask the subject to solve it in mental arithmetic. More specifically, the numerical calculation used in the experiments is the addition problem with the following forms: "augend + addend = sum" or "augend, summand 1, summand 2, summand 3, summand 4, summand 5 = sum". We can see that the former is a general addition problem, and the latter is a continuous addition problem. As for the continuous addition problem, the "+" symbol is not displayed, and the human subject adds up the numbers continuously.

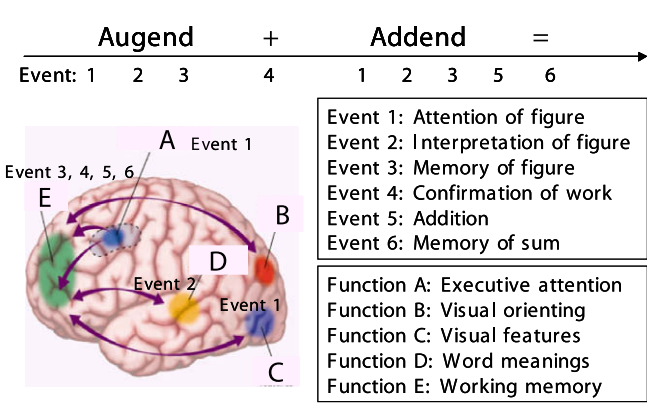
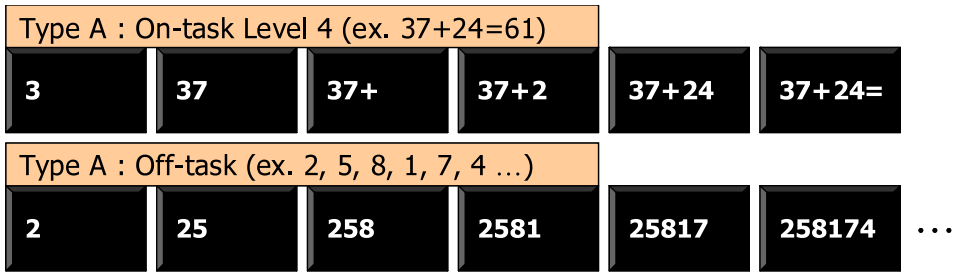


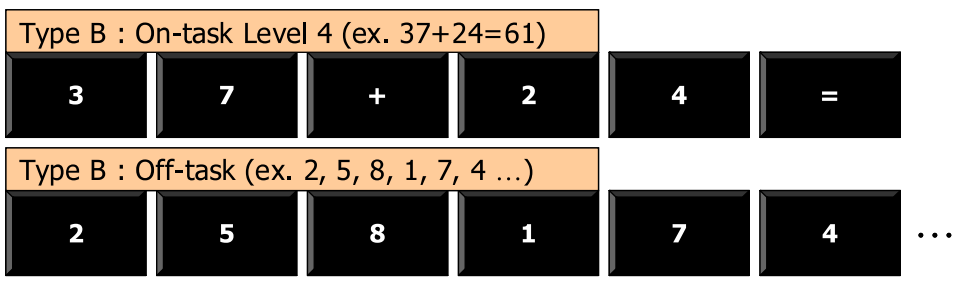
Fig. 2. Human computation processing.

In the experiments, four states (tasks), namely, *on-task*, *off-task*, *no-task*, and *memory-task* are addressed by the difference in the stimulus given to a human subject. Here *on-task* is the state on which the human subject is calculating by looking at a number; *off-task* is the state on which the human subject is looking at the number that appears at random; *no-task* is the relaxed state on which the human subject does not work at all; *memory-task* is the state on which the human subject is remembering six random numbers.

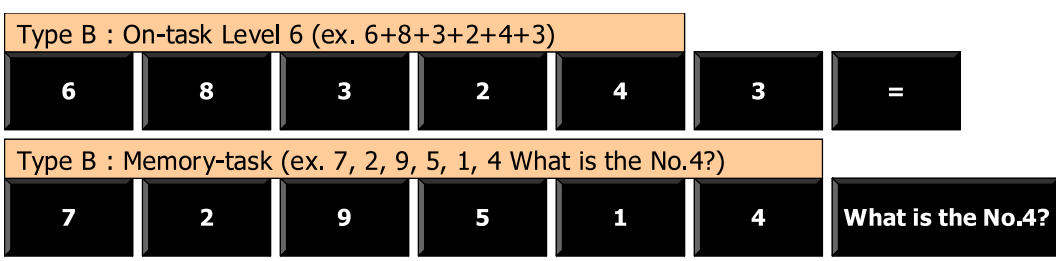
We set two presentation types, namely Type A and Type B, with six difficulty levels. Type A remains the figure on the screen, but Type B does not remain. Furthermore, the six difficulty levels are described as follows: Levels 1 and 2 use single digit addition tasks (with carry or no carry, respectively); Levels 3 and 4 use double digits addition tasks (with carry or no carry, respectively); Levels 5 and 6 use the tasks of 6 figures continuous addition. The difference between Level 5 and Level 6 is that the number of addition is 3 or less, and an answer is 11 or less in Level 5, however, carrying is caused two times in Level 6. Figs. 3 and 4 give examples of the screen state transition of the tasks, and show the relationship between tasks with respect to different types and difficulty levels. By such a design, it is possible to analyze the influence in the different difficulty levels, with the same *on-task* and *off-task*, and to make a



(a) Screen state of Type A (on-task and off-task)



(b) Screen state of Type B (on-task and off-task)



(c) Screen state of continuous addition and memory tasks

Fig. 3. Experimental design.

comparison between *on-task* and *off-task* in the same difficulty level.

3.2. Trigger signal and timing diagram

It is necessary to measure EEG/fMRI relevant to a certain event to the regular timing in measurement of ERP/fMRI repeatedly. In this research, since the attention was paid to each event of augend, "+", addend, and "=" presentations in computation activities. Pre-trigger was set to 200 ms, and addition between two digits is recorded in 2000 ms, respectively. Fig. 5 gives an example of the timing diagram, in which "Au" denotes an augend, "Ad" denotes an addend, "Sm" denotes a summand, and "Rn" denotes a random number. Hence "Au 1" is the most significant digit (MSD) of augend, and "Au 2" is the least significant digit (LSD) of augend. In this research, all tasks were executed in the EEG experiments. While only *on-task* in Level 4 was executed to confirm the effectiveness of the experiment design in the fMRI experiment. Subjects need to perform 25 experimental trials in EEG and 40 experimental trials in fMRI, respectively.

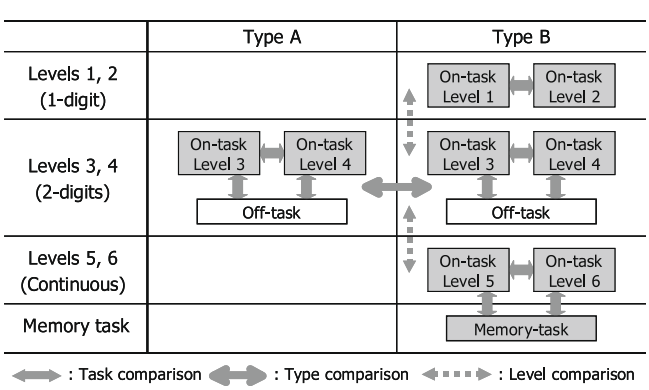


Fig. 4. An overview of experimental tasks with multiple difficulty levels and inter-relationships.

3.3. Experimental devices and measurement conditions

We first describe the EEG measurement conditions. The stimulus presentations were created using Presentation version 9.9 (Neurobehavioral) and the subjects were shown with the LCD. It was made to synchronize with PC for

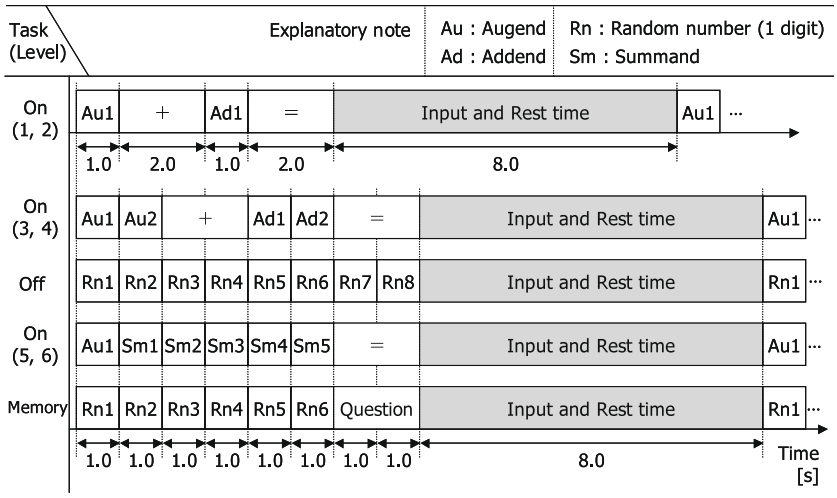


Fig. 5. Timing diagram of experimental tasks.

EEG data measurement with the trigger signal from PC for stimulus presentation control. The EEG activity was recorded using a 128 channel BrainAmp amplifier (Brain Products, Munich, Germany) with an 128 electrode cap as shown in Fig. 6. The impedance of the electrode is all 8 kΩ or less. The electrode cap is based on an extended international 10–20 system. Furthermore, the eye movement measurement (2 ch) is also used. The sampling frequency is 1000 Hz to be processed.

The ERP (event-related potential) data obtained from 10 subjects are analyzed using Brain Vision Analyzer (BRAIN PRODUCTS, Inc.) as the pre-processing and data analysis software of EEG. We used several functions provided in Brain Vision Analyzer, such as ocular correction, filter (BPF: 0.5–80 Hz), baseline correction, artifact rejection, and ERP derivation.

The fMRI measurement conditions are as follows. The stimulus presentations were also created using Presentation

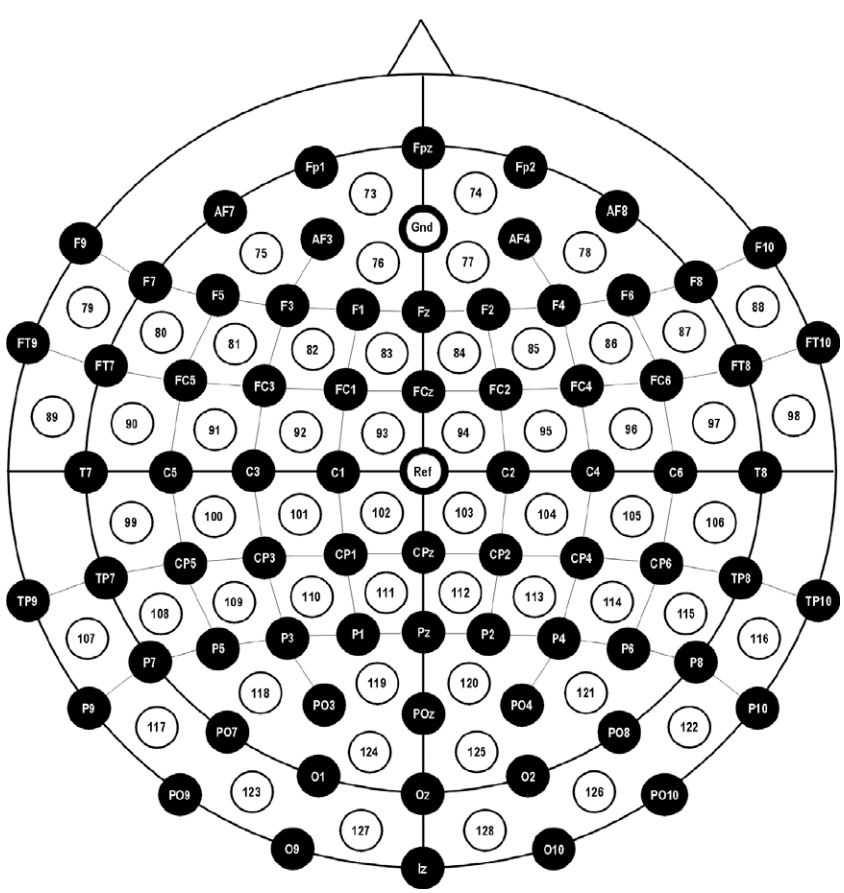


Fig. 6. Electrode cap.

version 9 (Neurobehavioral) and the subjects were shown with the LCD (viewed through a mirror). Scanning was performed on a 3 T Siemens system using a standard whole-head coil. Functional data were acquired using a gradient echo planar pulse sequence (TR=2 s, TE=31 ms, 30 axial slices, 4 mm^3 voxels, 0.8 mm inter-slice gap, 200 volumes per run, 90° flip angle, 64 × 64 matrix size in 240 × 240 mm^2 field of view). The imaging sequence is optimized to detect the BOLD effect including local shimming and 10 s of scanning prior to data collection to allow the MR signal to reach equilibrium.

The fMRI data obtained from 11 subjects are analyzed using Statistical Parametric Mapping (SPM2). We used several functions provided in SPM2, such as slice timing correction, realignment, coregistration, normalization, smoothing. A permissible threshold value of head movement is 2 mm, and data with a small movement is adopted. For normalization, the fMRI data are standardized with structural image and MNI standard brain of T1, and the parameter is set to T2-EPI. For smoothing, the FWHM is set to 10, which becomes two or three times of the voxel size. The activations reported survived an uncorrected voxel-level intensity threshold of $p < 0.01$.

4. Multi-aspect data analysis

It is possible that our experiments meet the following requirements: investigating the mechanisms of human visual and auditory systems, computation, problem-solving (Newell & Simon, 1972), and the spatiotemporal characteristics and flow of HIPS in general. Furthermore, multi-aspect data analysis can mine several kinds of interesting patterns and hypotheses from different data sources. In this research, we used multiple data sources, such as frequency spectrum data, ERP data, potential topography (2D and 3D), and fMRI data, for multi-aspect data analysis, namely frequency analysis, ERP analysis, potential topography analysis, and fMRI data analysis. Furthermore, a more exact result can be obtained by integrating and explaining the results of multi-aspect data analysis.

4.1. Frequency analysis

The objective of frequency analysis is to catch the frequency element fluctuation from the spatiotemporal view by recognizing distribution and appearance of alpha wave and beta wave. Hence, we can find the different features on frequency elements with respect to a computation processing. And medical knowledge can be also used by analyzing frequency elements.

Fig. 7 shows the results of frequency analysis with respect to no-task, off-task, and on-task (Level 4), respectively. We analyzed the data (about 1 s) when the addend (Ad 2) was presented in on-task. The time period is closely related to a computation activity. Furthermore, we analyzed the data (about 1 s) when the Rn 5 was presented in off-task. In this work, we focus on analyzing the following channels: Fz (attentiveness and concentration), Oz (visual cortex), C5 (left temporal lobe), and CPz (between Fz and Oz). In comparison between on-task and off-task, we can see that the brain wave has a high percentage of theta rhythm in Fz and Oz in on-task. On the other hand, the alfa rhythm pre-

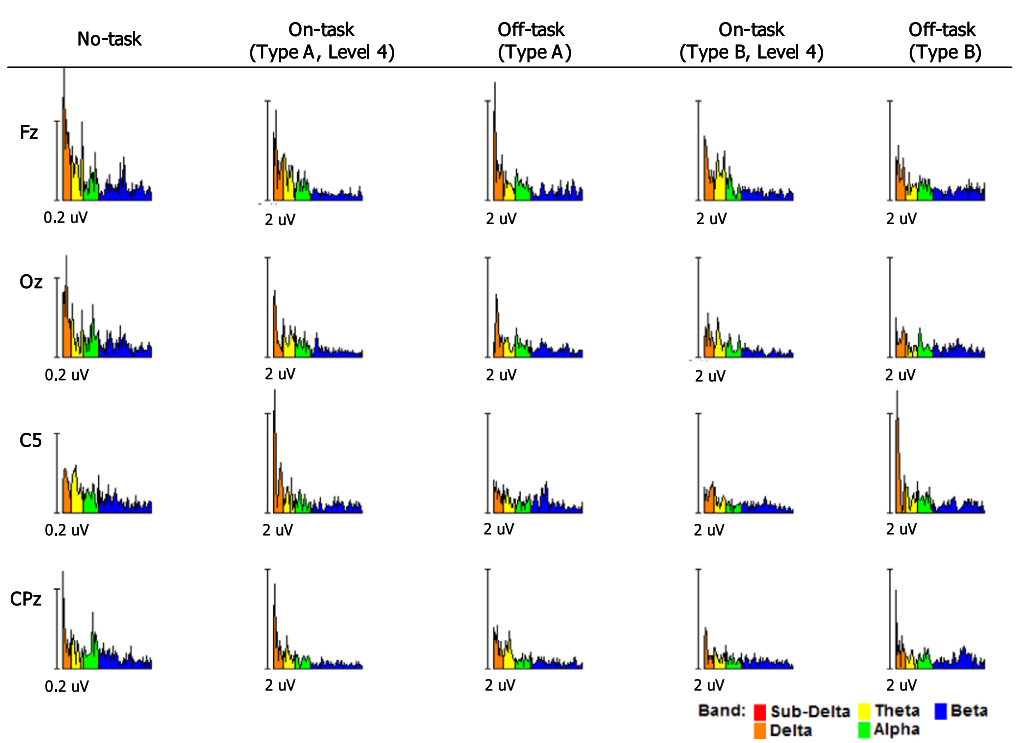


Fig. 7. Frequency band of brain waves.

dominates in no-task, and decreases in on-task and off-task. This phenomenon is called the Fm theta rhythm, which is a conventional knowledge confirmed.

4.2. ERP data analysis

For the measured EEG data, a maximum of 40 addition average processing was performed, and the ERP (event-related potential) human brain waves were derived by using Brain Vision Analyzer (Brain Products, Munich, Germany). ERP is a light, sound, and brain potential produced with respect to the specific phenomenon of spontaneous movement (Handy, 2004; Nittono, Nageishi, Nakajima, & Ullsperger, 1999; Picton et al., 2000). We use the mental arithmetic task with visual stimuli, as an example to investigate human computation processing.

Generally speaking, the Wernicke area of a left temporal lobe and the prefrontal area are related to the computation processing (Mizuhara, Wang, Kobayashi, & Yamaguchi, 2005). In this research, we pay attention to integrated processing and short-term memory (AF3), recognition of the number (CP3), vision (Oz), as well as compare between tasks by focusing on some important channels.

Fig. 8a shows the ERP of Type A on-task in Level 4 and off-task. In comparison between on-task and off-task in Level 4, the latency of response is slow in on-task. This phenomenon is especially prominent in Oz during the "+" presented, AF3 and CP3 during an addend presented, and Oz during the "=" presented. The most interesting difference is about a latent time during the Ad 2 presented. P250 appears in off-task and P300 appears in on-task on AF3. Similarly, N150 and P300 appear in off-task, but P400 (long latency potential) appears in on-task on CP3. We can guess that the delay of AF3 and CP3 is related to the difference of digit number and computation activities. This result suggests that the figures gaze and linguistic interpretation possess a lower speed. On the other hand, the positive potential of on-task is higher than that of off-task after an addend was presented. It is thought that this result is related to a peculiar process of on-task like the addition processing and the short-term memory.

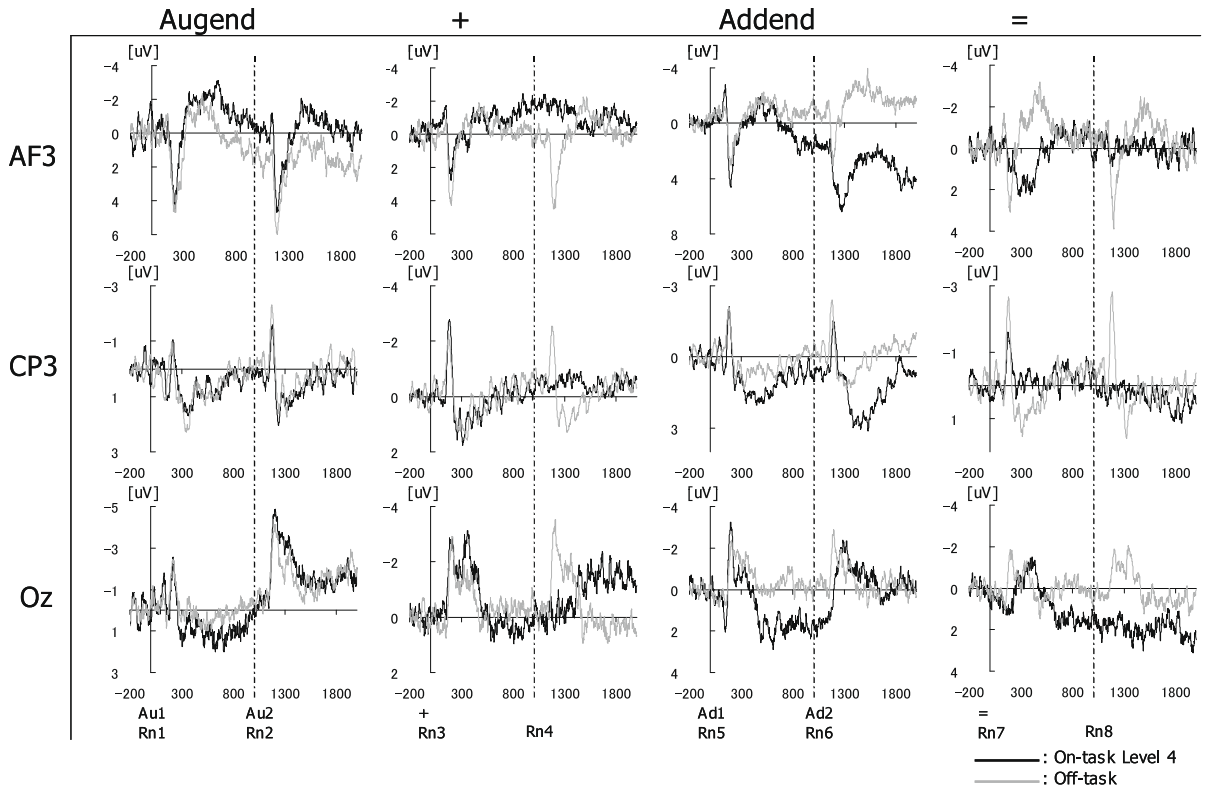
Fig. 8b shows the ERP of Type B on-task in Level 6, memory-task and off-task. A common factor to the three tasks is that the figure (1 digit) is continuously displayed on the monitor. On the contrary, the differences among the three tasks are with respect to adding, memory, and attention. In comparison between on-task and memory-task, the potential change of on-task is bigger while the figure appears. On the other hand, the potential change of memory-task is very big while the question time. In particular, the latent time of P250 is deeply in AF3. We can guess that the potential change is related to the process of short-term matching (Fletcher & Henson, 2001). As in Type A, both on-task and memory-task latent times are slower than that of off-task. The N300 and N400 can be explained as language understanding and semantic process (Curran, Tucker, Kutas, & Posner, 1993). We need to discuss deeply by analyzing the difference of the appearance of N400 in the different levels of computation difficulty.

4.3. Potential topography analysis

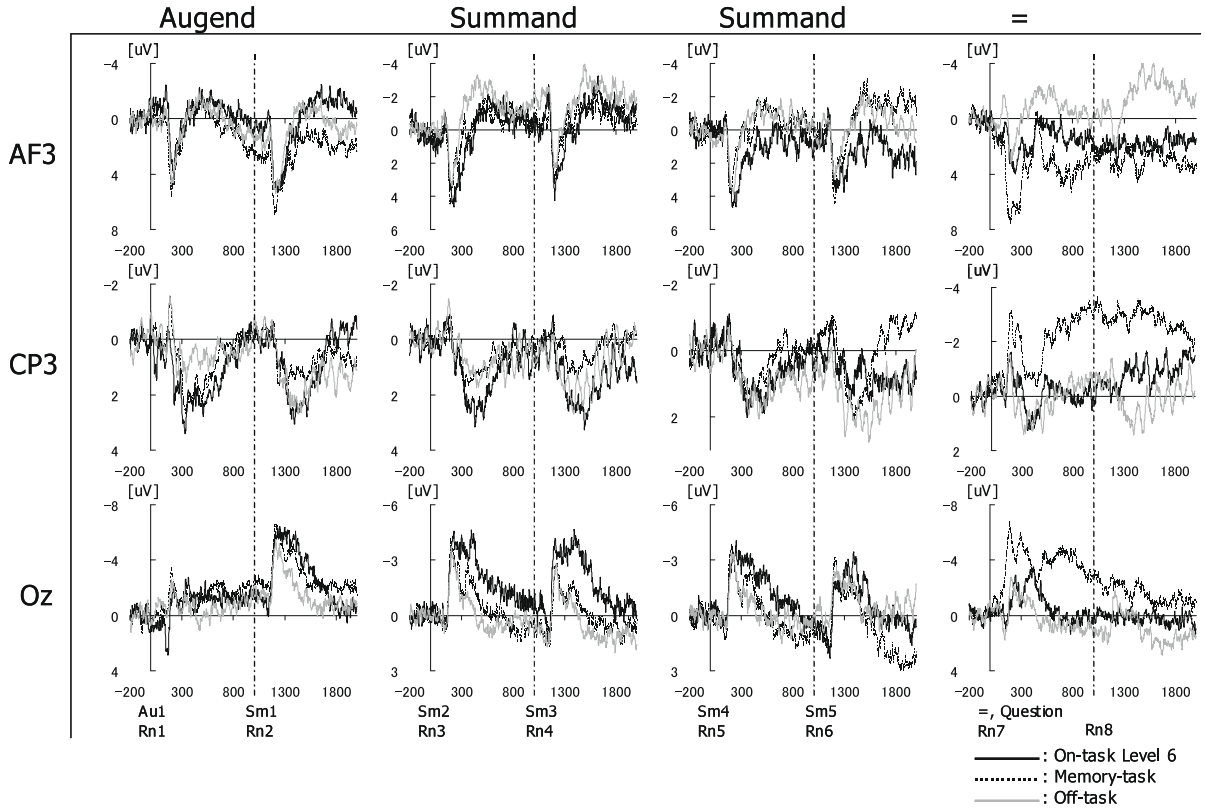
The advantage of potential topograph analysis is that it catches the electrical fluctuation and gives a rough visualization of the spatiotemporal characteristics, for recognizing distribution and appearance of positive and negative potentials. Hence, we can find the different features on potentials with respect to each step of the computation processing. In this research, we use the 2D and 3D topographies. In particular, activities around the brain become clear by using the 3D topography (Kim, Kwon, Lee, & Kim, 2007).

Fig. 9a shows the 2D potential topography. This figure shows the appearance of the potential change from beginning to the end of a computation processing. Each figure is a description between 300 ms and 400 ms from the stimulation presentation. In this research, we focus on on-task in Level 4 (both Type A and Type B), off-task, and memory-task. In comparison between four tasks, it is found that the appearance position and timing of positive potential are greatly different. In comparison between Type A and Type B, the positive potential distribution of Ad 1 and Ad 2 is almost symmetrical in Type A, but it is not symmetry (right brain-dominant) in Type B. >From the viewpoint of cerebral localization, this result is very important for logical or intuitive computation processing. Furthermore, in comparison between on-task and off-task, the potential distributions of frontal cortex and parietal lobe are sharply different. The positive potential just appears in the frontal cortex from the parietal lobe in memory-task and on-task, but such phenomenon does not appear in the visual cortex in off-task.

Fig. 9b shows the 3D potential topography, in which we can analyze activations of the brain including the left temporal lobe with respect to a computation processing that cannot be observed by the 2D topography. Hence, we focus on the appearance time of an addend which is most deeply related to the computation, and we describe the potential change from the left side with three-dimensionally viewpoint in 200 ms. In this research, we focus on on-task in Level 4 (both Type A and Type B), on-task in Level 6 (Type B), off-task and memory-task. In comparison between Type A and Type B in Level 4, we can find some common patterns with respect to the positive potential, and the same changes of the potential in the left temporal lobe and the Wernicke area. On the other hand, the potential change of the visual area is considerably different. This result suggests that subjects keep a careful gaze at figures in Type B. As for an off-task, the positive potential is not observed in the frontal cortex by using both 2D and 3D topographies. However, we can guess, from the activations of the visual and Wernicke areas, that the subjects are recognizing what number appeared. As for a memory-task, the positive potential is not observed in the Wernicke area. This phenomenon suggests that memorizing numbers is not

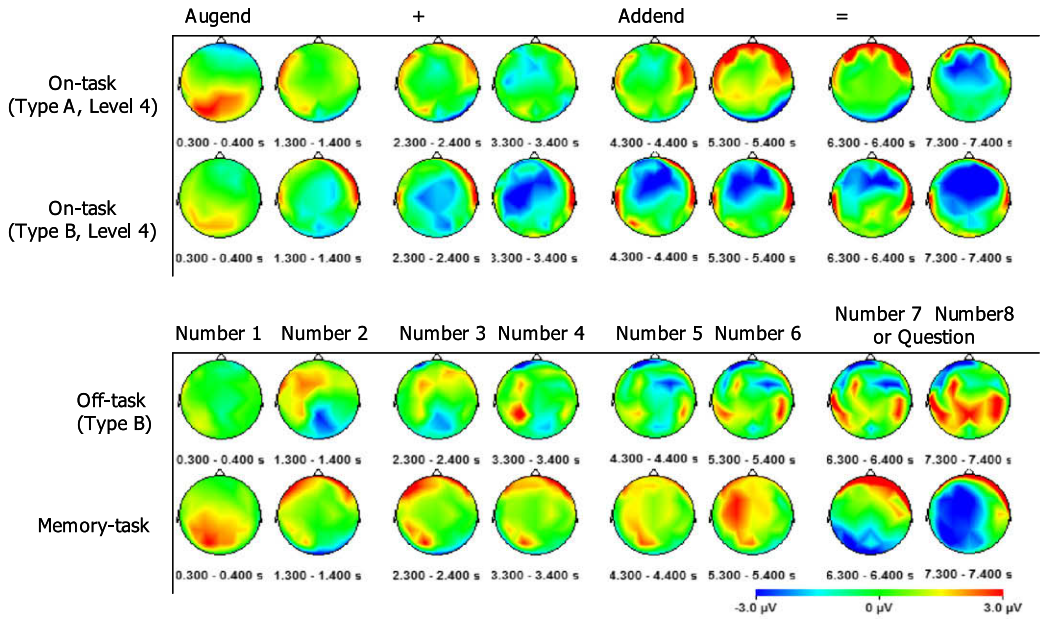


(a) ERP of Type A (on-task in Level 4 and off-task)

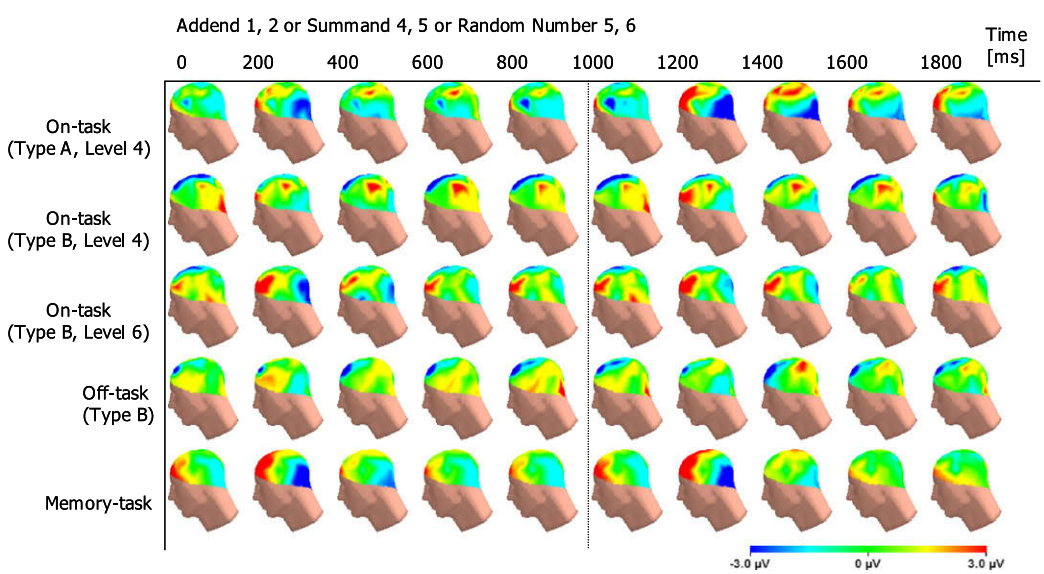


(b) ERP of Type B (on-task in Level 6, memory-task and off-task)

Fig. 8. ERP data analysis.



(a) 2-D potential topography of tasks



(b) 3-D potential topography of tasks

Fig. 9. Topograph analysis.

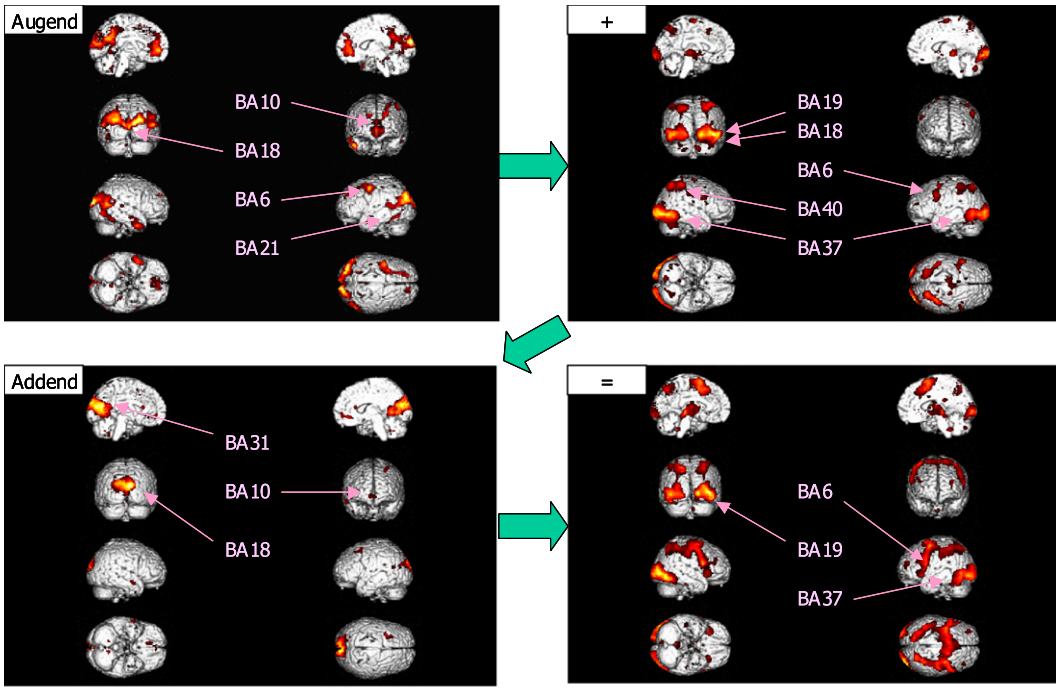
solved as a memory-task by linguistic interpretation of numbers, but as symbols. The results show the differences among attention, recognition, and memory processes in each task from the macro viewpoint of brain functions for the computation processing. However, such EEG topographies have limitation in spatial resolution, and need to combine with fMRI data analysis for identifying the activated areas more accurately.

4.4. fMRI data analysis

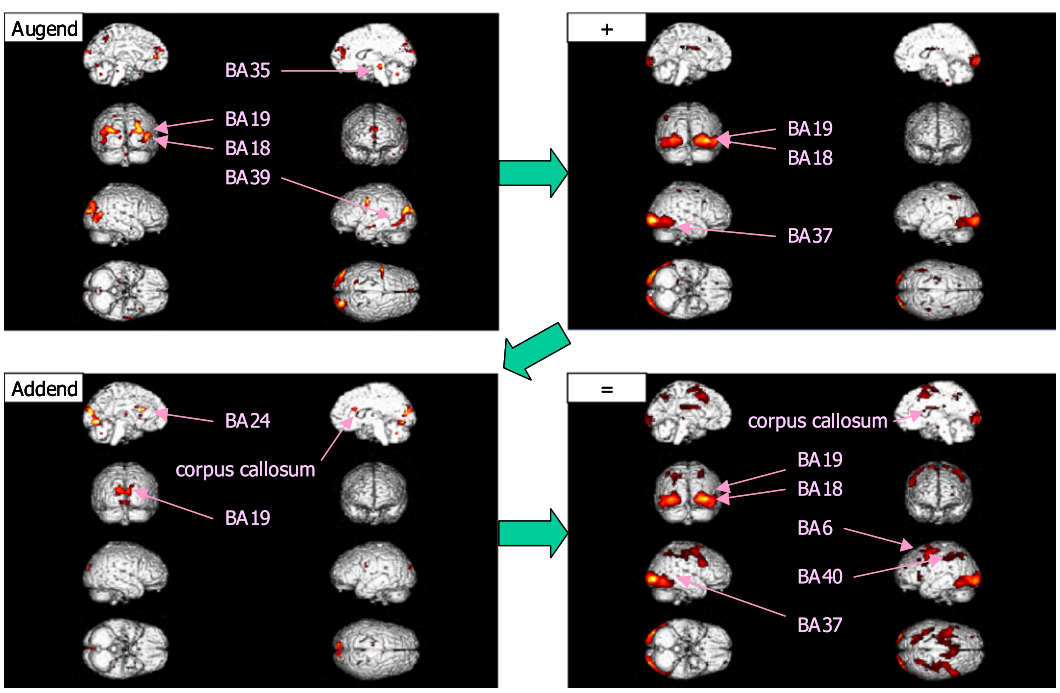
fMRI makes the observation of the brain more accurate and detailed by providing images of functional brain activations to observe dynamic activation patterns within different areas of the brain for a given task (Rosen, Buckner, & Dale, 1998; Shulman & Rothman, 1998). It is excellent in the space resolution, but inferior in the time resolution.

Fig. 10 shows results of fMRI data group analysis in on-task Type A in Level 4 (Fig. 10a) and on-task Type B in Level 4 (Fig. 10b), respectively. The results demonstrate how the activated areas of the brain operate at a specific point of time, and how the operations change over time when subjects carried out the computation processing.

As shown in Fig. 10a, we advocate the hypotheses concerning the computation processing in Type A according to



(a) fMRI data analysis in Type A (Level 4)



(b) fMRI data analysis in Type B (Level 4)

Fig. 10. fMRI data analysis.

the results and well-known knowledge of Brodmann areas. First, the subject received visual information (BA 18) when an augend was presented. Next, the subject integrated visual information (BA 6) and gave the linguistic interpretation in the Wernicke area (BA 40) when the "+" symbol was presented. The subject received visual information (BA 18) and evoked a memory (BA 31) when an addend was presented. Finally, the subject carried out the computation processing (BA 6) when the "=" symbol was presented.

Similar to Type A, we advocate the hypotheses concerning the computation processing in Type B as shown in Fig. 10b. First, when an augend was presented, the subject received visual information (BA 18). Next, the subject memorize visual information (BA 37) when the "+" sym-

	Type A	Type B
Frequency Analysis	Delta rhythm decreased in the visual cortex	Delta rhythm decreased in the visual cortex
ERP Analysis	P250 appeared in the frontal cortex	Similar to Type A, but P250 is indistinctness
Topograph Analysis	Positive potential shifted from CP3 to T7 and FT7	Positive potential shifted from CP3 to T7 and T8
fMRI Analysis	Activated areas : BA 18, BA 21, BA 10	Activated areas : BA 19, BA 18, BA 35
	Augend 1 ⇨ Augend 2	Augend 1 ⇨ Augend 2

(a) Comparing Type A and Type B (augend)

	Type A	Type B
Frequency Analysis	Beta rhythm decreased in the parietal lobe	Beta rhythm decreased in the parietal lobe
ERP Analysis	N400 appeared in the visual cortex Oz	N250 appeared in the visual cortex Oz
Topograph Analysis	Positive potential shifted from CP3 to FT7	Negative potential appeared in the parietal cortex
fMRI Analysis	Activated areas : BA 19, BA 37, BA 40, BA 6	Activated areas : BA 18, BA 19, BA 37
	+	+

(b) Comparing Type A and Type B (+)

	Type A	Type B
Frequency Analysis	Theta rhythm increased in the Fz channel	Theta rhythm increased in the Fz channel
ERP Analysis	P300-P400 appeared (Addend 2 presentation)	P500 (long-latency) appeared in the P3
Topograph Analysis	Positive potential appeared in the Pz, frontal cortex	Positive potential appeared in the P5, C5, F8
fMRI Analysis	Activated areas : BA 18, BA 31, BA 10	Activated areas : BA 19, BA 24, corpus callosum
	Addend 1 ⇨ Addend 2	Addend 1 ⇨ Addend 2

(c) Comparing Type A and Type B (addend)

	Type A	Type B
Frequency Analysis	Theta rhythm appeared in a higher percentage	Theta rhythm appeared in a higher percentage
ERP Analysis	Latent time is not clear in the Wernicke area	Latent time is not clear in the Wernicke area
Topograph Analysis	Positive potential appeared in the FPz	Positive potential appeared in the P7, FT8
fMRI Analysis	Activated areas : BA 6, BA 37, BA 19	Activated areas : BA 6, BA 40, corpus callosum
	=	=

(d) Comparing Type A and Type B (=)

Fig. 11. Integration and explanation of Type A and Type B.

bol was presented. The subject received visual information (BA 19) and evoked a memory (BA 24) when an addend was presented. Finally, the subject gave the linguistic interpretation in the Wernicke area (BA 40) and carried out the computation processing (BA 6) when the "=" symbol was presented.

From the results, we can see that some different brain areas are activated over the time when using different presentation methods. This phenomenon is also found from the EEG/ERP results, and it is related to the adaptability of human computation mechanism. The subjects can view tasks of the computation problem many times, since the figures remain on the screen in Type A. Hence, we can guess that the subjects compute repeatedly. On the other hand, the activated areas that relate to perception, memory, and attention stand out in the first half of the result in Type B. >From the second half of the result in Type B, we can see that the activation in the corpus callosum is with a high probability, which means that the subject processed a figure by coordinating between the right brain and the left brain.

5. Integration and explanation of results

Based on the results stated above, this section provides a comparative study by combining the results obtained by each analysis method separately towards a more whole explanation and understanding. The common features and the difference points are investigated by comparing Type A with tasks of Type B in Level 4, which are the two digit addition problem. However, the repetition problem related tasks, such as off-task, continuous addition task

(Level 6), and memory-task, cannot be indiscriminately discussed on the same axis as the two digit addition problem. Hence, the comparison between these tasks is summarized by only discussing some interesting points.

We first discuss the difference of the presentation methods between Type A and Type B at the same level (on-task in Level 4). Fig. 11 shows the main features of Type A and Type B on each event. >From the results of brain wave analysis and fMRI data analysis, we can see that timing of figure recognition is different from that of computation. Furthermore, since subjects must reassemble two digits figure based on two characters in Type B, we can confirm from the result of fMRI data analysis in Type B that the brain areas with respect to restructuring and working memory of the number are activated (BA 6, BA 37, BA 40). We also find the spatiotemporal difference that the positive potential appears in Type A and Type B from the result of topography. The positive potential in Type A stands out in each time zone, and this suggests the continuousness of the computation processing. On the other hand, a common feature of Type A and Type B is the spatial recognition for the figure, which is obtained from both fMRI data analysis (BA 7) and brain wave analysis (CP 3). An interesting guess from the results is that human beings can do an efficient computation processing unconsciously and the difference of processing affects the difference of the results. In order to understand the results deeply, not only brain data analysis, but also the analyzes need to be combined with studies of psychology and pedagogy.

Next, we give a summary by discussing some interesting points in comparison between tasks in Type B. It is clear from the results that both common potential change and peculiar potential change existed. The results suggest that the computation processing consists of more component functions, and both on-task and off-task use the same brain functions for figure recognition. A lot of common features were also confirmed from the visual cortex to the left temporal lobe in this analysis. On the other hand, comparing with a usual addition problem, it was found that the frontal lobe was highly activated when subjects carried out the continuous addition problem that puts subjects a high load. Although the difference in the figure recognition is clear in the memory-task, it is not caught plainly that the reactive difference between the right and left brain when carried out different tasks For such a further investigation, it is necessary to compare the computation problem with other higher cognitive functions such as reasoning (Liang et al., 2007), which is based on BI methodology for systematic experimental design and analysis.

6. Conclusion

In this paper, we investigated human computation mechanism related functions by using mental arithmetic as an example, which demonstrate what is BI methodology and its usefulness. The proposed multi-aspect data analysis approach based on BI methodology shifts the focus of cognitive science from a single type of experimental data analysis towards a deep, holistic understanding of human information processing principles, models and mechanisms. Based on the BI methodology, we designed cognitive experiments for investigating human computation mechanism with multiple difficulty levels on multi-task, carried out the experiments by using fMRI and EEG, and analyzed the data from multi-aspect. Experimental results showed the usefulness of our approach.

Our future work includes obtaining and analyzing more subject data, and apply more data mining techniques to richen multi-aspect data analysis (Motomura, Hara, Zhong, & Lu, 2007;Zhong, 2007). In addition, the computation processing can be regarded as an example of problem-solving processes, and can be deeply investigated in the connection between the figure processing and memory, as well as the relationship with other higher cognitive functions.

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References

Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. (2004). An integrated theory of mind. *Psychological Review, 111*, 1036–1060.

Anderson, J. R., Qin, Y., Yung, K.-J., & Carter, C. S. (2007). Information-processing modules and their relative modality specificity. *Cognitive Psychology, 54*, 185–217.

Curran, T., Tucker, D. M., Kutas, M., & Posner, M. I. (1993). Topography of the N400: Brain electrical activity reflecting semantic expectancy. *Electroencephalography and Clinical Neurophysiology, 88*, 188–209.

Fletcher, P. C., & Henson, R. N. A. (2001). Frontal lobes and human memory insights from functional neuroimaging. *Brain, 124*, 849–881.

Gazzaniga, M. S. (Ed.). (2004). *The cognitive neurosciences III*. The MIT Press.

Handy, T. C. (2004). *Event-related potentials, a methods handbook*. The MIT Press.

Kim, K. H., Kwon, J. H., Lee, D. H., & Kim, S. I. (2007). The 3D brain topography based on PC, engineering in medicine and biology society. In *Proceedings of the 19th annual international conference of the IEEE* (Vol. 4, pp. 1599–1601).

Liang, P., Zhong, N., Wu, J. L., Lu, S., Liu, J., & Yao, Y. Y. (2007). Time dissociative characteristics of numerical inductive reasoning: Behavioral and ERP evidence. In *Proceedings of 2007 international joint conference on neural networks (IJCNN'07)*. IEEE Press.

Megalooikonomou, V., & Herskovits, E. H. (2001). Mining structure-function associations in a brain image database. In K. J. Cios (Ed.), *Medical data mining and knowledge discovery* (pp. 153–179). Physica-Verlag.

Mitchell, T. M., Hutchinson, R., Niculescu, R. S., Pereira, F., Wang, X., Just, M., et al. (2004). Learning to decode cognitive states from brain images. *Machine Learning, 57*(1–2), 145–175.

Mizuhara, H., Wang, L., Kobayashi, K., & Yamaguchi, Y. (2005). Long-range EEG phase-synchronization during an arithmetic task indexes a coherent cortical network simultaneously measured by fMRI. *NeuroImage, 27*(3), 553–563.

Motomura, S., Hara, A., Zhong, N., & Lu, S. (2007). An investigation of human problem solving system: Computation as an example. In M. Kryszkiewicz et al. (Eds.), *Rough sets and intelligent systems paradigms*. *LNAI 4585* (pp. 824–834). Springer.

Newell, A., & Simon, H. A. (1972). *Human problem solving*. Englewood Cliffs: Prentice-Hall.

Nittono, H., Nageishi, Y., Nakajima, Y., & Ullsperger, P. (1999). Event-related potential correlates of individual differences in working memory capacity. *Psychophysiology, 36*, 745–754.

Picton, T. W., Bentin, S., Berg, P., Donchin, E., Hillyard, S. A., Johnson, R., et al. (2000). Guidelines for using human event-related potentials to study cognition, recording standards and publication criteria. *Psychophysiology, 37*, 127–152.

Rosen, B. R., Buckner, R. L., & Dale, A. M. (1998). Event-related functional MRI: Past, present, and future. *Proceedings of National Academy of Sciences, USA, 95*(3), 773–780.

Shulman, R. G., & Rothman, D. L. (1998). Interpreting functional imaging studies in terms of neurotransmitter cycling. *Proceedings of National Academy of Sciences, USA, 95*(20), 11993–11998.

Sommer, F. T., & Wichert, A. (Eds.). (2003). *Exploratory analysis and data modeling in functional neuroimaging*. The MIT Press.

Sternberg, R. J., Lautrey, J., & Lubart, T. I. (2003). *Models of intelligence*. American Psychological Association.

Zhong, N. (2005). Building a brain-informatics portal on the wisdom web with a multi-layer grid: A new challenge for web intelligence research. In V. Torra et al. (Eds.), *Modeling decisions for artificial intelligence*. *LNAI 3558* (pp. 24–35). Springer.

Zhong, N. (2006). *Impending brain informatics (BI) research from web intelligence (WI) perspective*. *International journal of information technology and decision making* (Vol. 5(4)). World Scientific (pp. 713–727).

Zhong, N. (2007). Actionable knowledge discovery: A brain informatics perspective, special trends and controversies department on domain-driven actionable knowledge discovery. *IEEE Intelligent Systems*, 85–86.

Zhong, N., Liu, J., Yao, Y. Y., Wu, J., Lu, S., & Li, K. (Eds.). (2007). *Web intelligence meets brain informatics*. *State-of-the-art-survey, LNCS 4845* (pp. 223–243). Springer.

Zhong, N., Wu, J. L., Nakamaru, A., Ohshima, M., & Mizuhara, H. (2004). *Peculiarity oriented fMRI brain data analysis for studying human multi-perception mechanism*. *Cognitive systems research* (Vol. 5(3)). Elsevier (pp. 241–256).